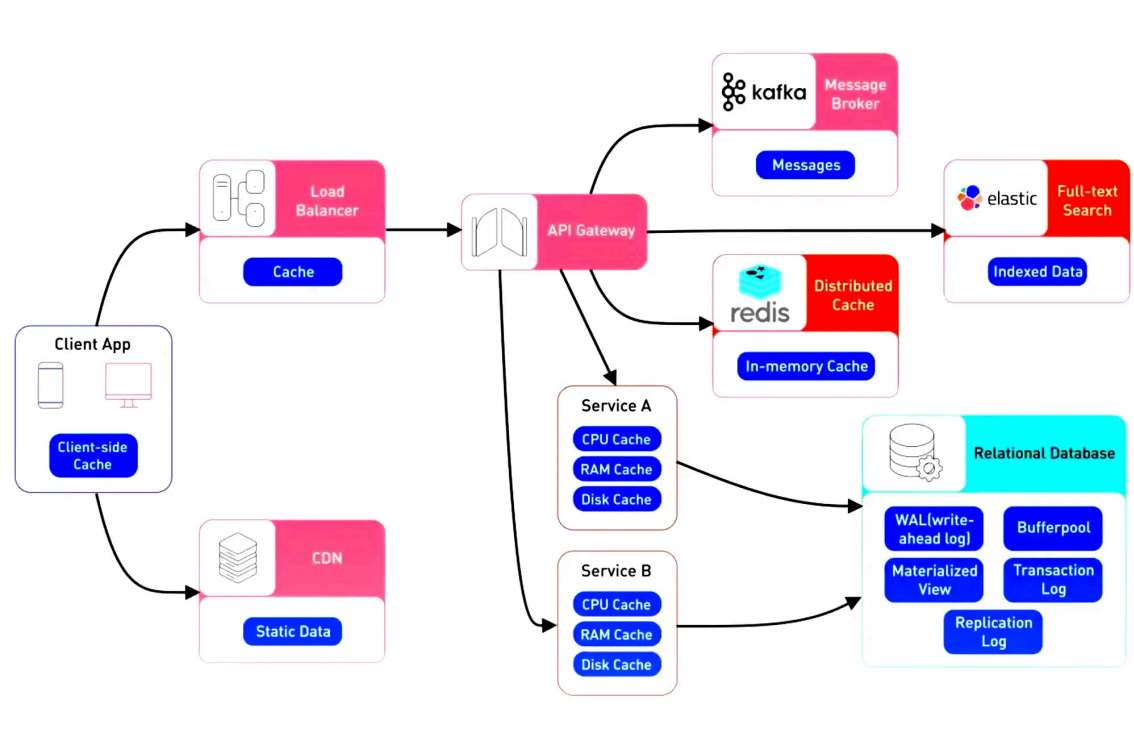
Distributed Caching in Microservices


Client App
Client-side Cache
Load Balancer
Cache
CDN
Static Data
API Gateway
kafka
Message Broker
Messages
redis
Distributed Cache
In-memory Cache
elastic
Full-text Search
Indexed Data
Service A
CPU Cache
RAM Cache
Disk Cache
Service B
CPU Cache
RAM Cache
Disk Cache
Relational Database
WAL(write-ahead log)
Bufferpool
Materialized View
Transaction Log
Replication Log

Caching stores frequently accessed data so microservices can serve it without repeating expensive operations. Benefits:

- **Boosts performance** – faster responses by avoiding repeated DB queries or heavy computations.
- **Enhances resiliency** – services can respond even if backend systems are temporarily unavailable.
- **Improves scalability** – reduces load on backend systems by offloading repeated requests.

Cache Patterns

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1. Cache-Aside Pattern (Application-Controlled)

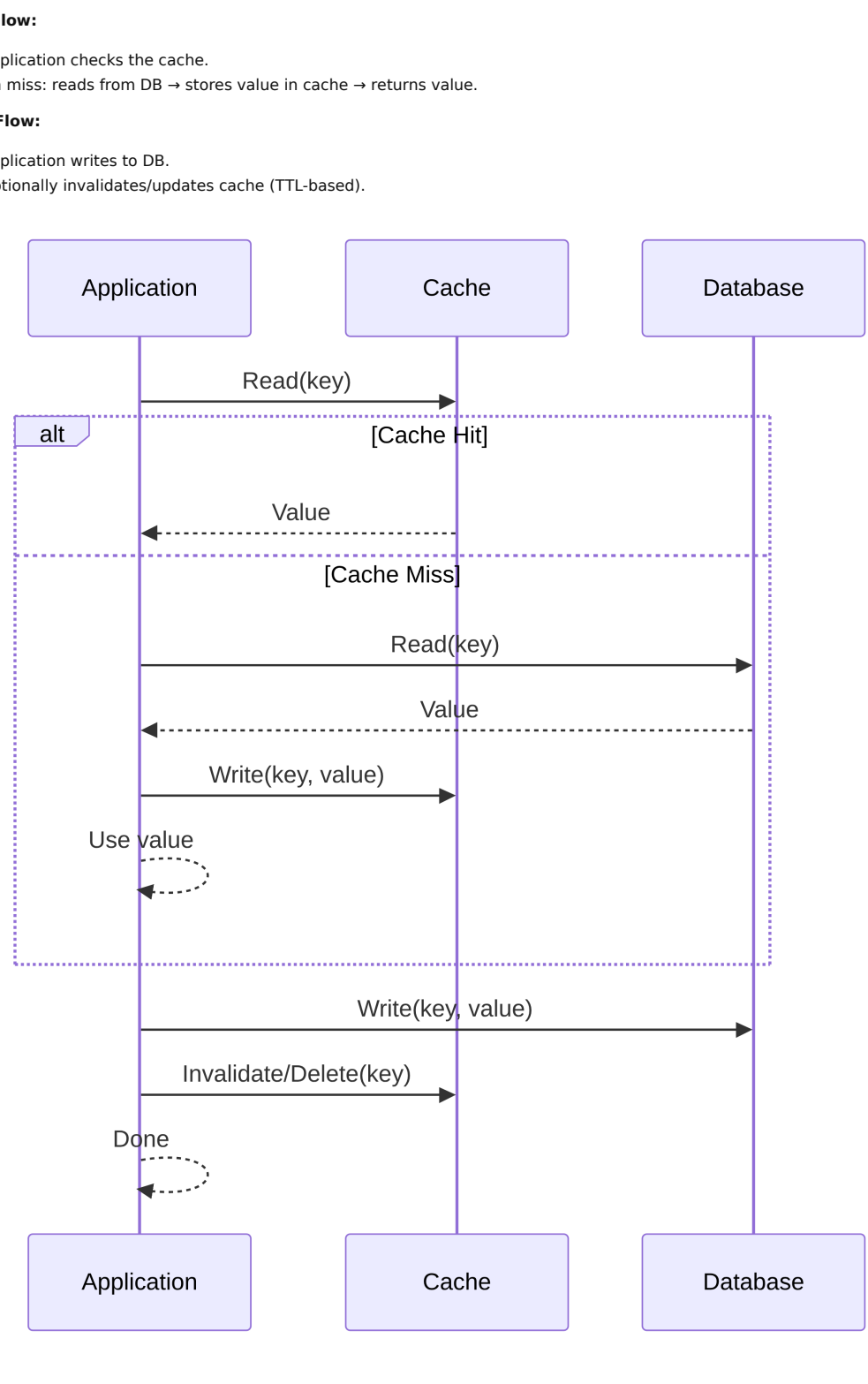
> *Very flexible and widely used. Suitable for read-heavy workloads. Application manages cache population.*

Read Flow:

- Application checks the cache.
- On miss: reads from DB → stores value in cache → returns value.

Write Flow:

- Application writes to DB.
- Optionally invalidates/updates cache (TTL-based).


Application
Cache
Database
Read(key)
alt
[Cache Hit]
Value
[Cache Miss]
Read(key)
Value
Write(key, value)
Use value
Write(key, value)
Invalidate/Delete(key)
Done
Application
Cache
Database

Pros: Full control, flexible, good for read-heavy workloads.

Cons: More complex, risk of stale cache, extra round-trips on miss.

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2. Read-Through Pattern

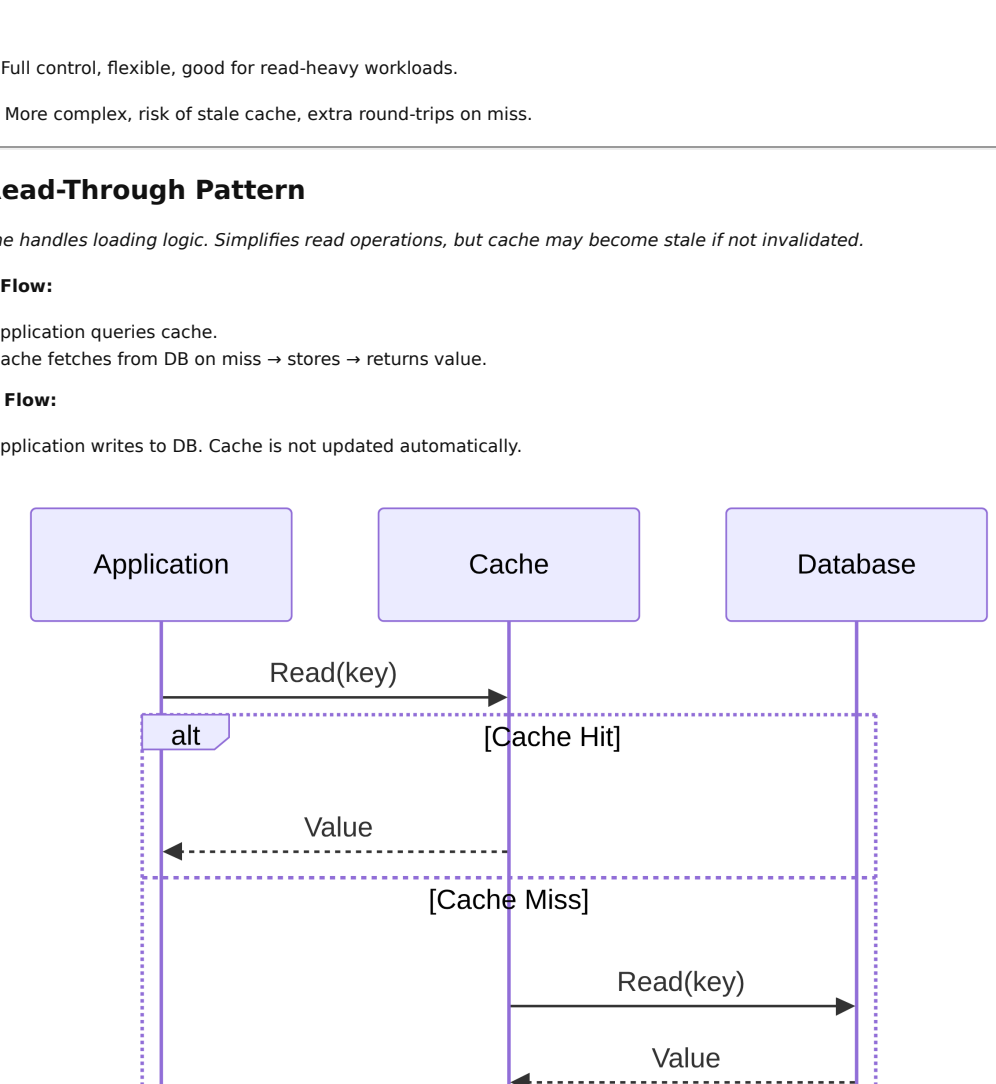
> *Cache handles loading logic. Simplifies read operations, but cache may become stale if not invalidated.*

Read Flow:

- Application queries cache.
- Cache fetches from DB on miss → stores → returns value.

Write Flow:

- Application writes to DB. Cache is not updated automatically.


Application
Cache
Database
Read(key)
alt
[Cache Hit]
Value
[Cache Miss]
Read(key)
Value
Value
Optionally store value (TTL / Eviction)
Write(key, value)
Application
Cache
Database

Pros: Simplifies client code.

Cons: Cache may serve stale data, requires cache logic to fetch from DB.

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3. Write-Through Pattern

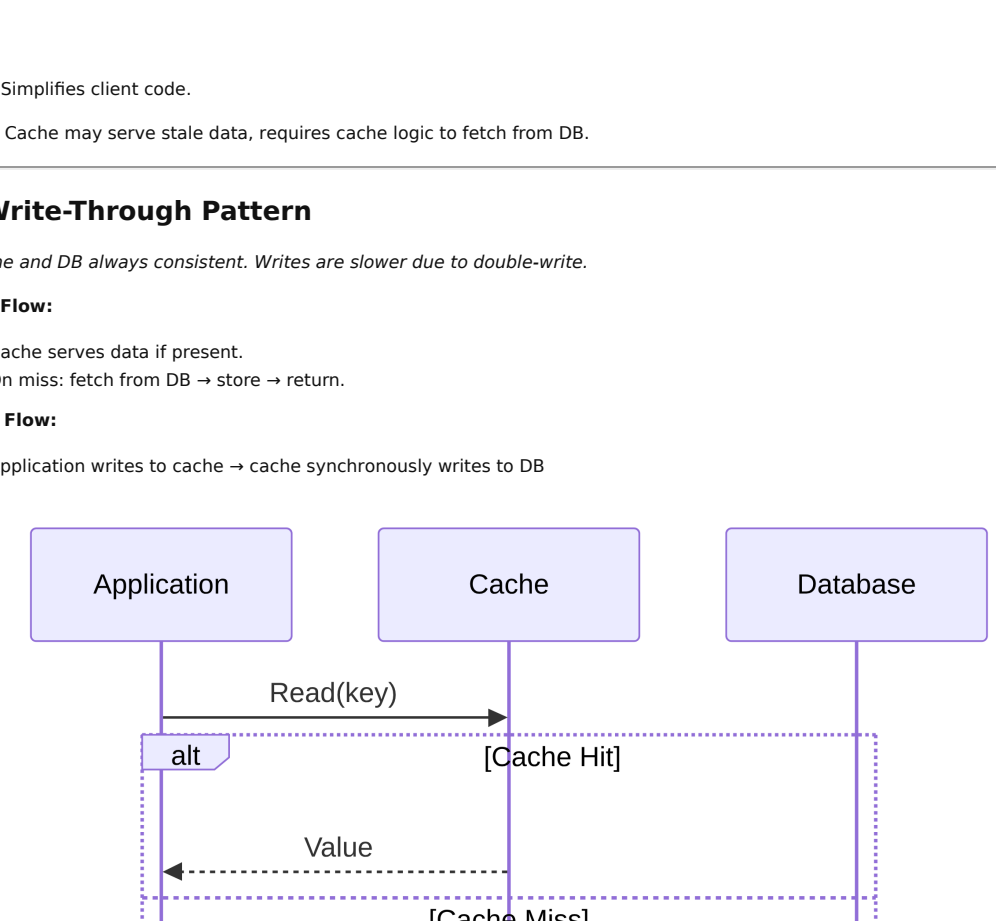
> *Cache and DB always consistent. Writes are slower due to double-write.*

Read Flow:

- Cache serves data if present.
- On miss: fetch from DB → store → return.

Write Flow:

- Application writes to cache → cache synchronously writes to DB


Application
Cache
Database
Read(key)
alt
[Cache Hit]
Value
[Cache Miss]
Read(key)
Value
Value
Write(key, value)
Write(key, value)
Done
Application
Cache
Database

Pros: Strong consistency.

Cons: Slower writes.

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4. Write-Behind / Write-Back Pattern

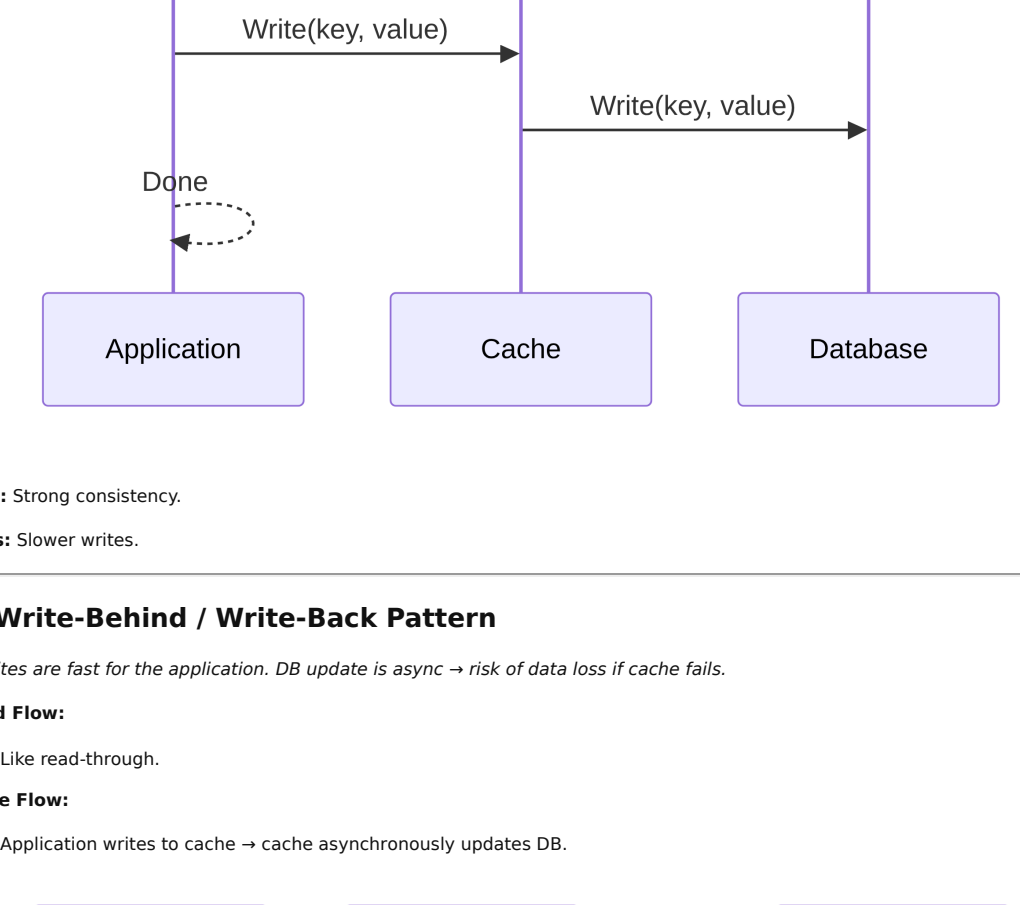
> *Writes are fast for the application. DB update is async → risk of data loss if cache fails.*

Read Flow:

- Like read-through.

Write Flow:

- Application writes to cache → cache asynchronously updates DB.


Application
Cache
Database
Read(key)
alt
[Cache Hit]
Value
[Cache Miss]
Read(key)
Value
Value
Write(key, value)
Cache updates in-memory value
DB update is async / delayed
(Eventually) Write(key, value)
Done
Application
Cache
Database

Pros: Very fast writes, good for high-throughput systems.

Cons: Potential data loss, complex async handling.

Cache Invalidation

> *Cache invalidation is a "hard problem" due to the tradeoff between performance (large, long-lived cache) and consistency (fresh data).*

Ensuring cached data remains accurate is challenging in distributed systems. Strategies:

- **Time-based**: Expire entries after a set period. Simple but may serve stale data.
- **Event-based**: Invalidate on data changes. Consistent but complex to implement.
- **Size-based**: Evict entries when cache exceeds memory limit. Memory-efficient, may remove relevant data.

Resources